



MATHEMATICS (EXTENSION 2)

2012 HSC Course Assessment Task 4

August 17, 2012

General instructions

- Working time – 50 min.
- **Commence each new question on a new page.**
- Write using blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- Board approved calculators may be used.
- All necessary working should be shown in every question. Marks may be deducted for illegible or incomplete working.
- Attempt **all** questions.
- At the conclusion of the examination, bundle the booklets used in the correct order within this paper and hand to examination supervisors.

Class (please ✓)

- ☐ 12M4A – Mr Lin
- ☐ 12M4B – Mr Ireland
- ☐ 12M4C – Mr Fletcher

STUDENT NUMBER # BOOKLETS USED:

Marker's use only.

QUESTION	1	2	3	Total	%
MARKS	$\overline{12}$	$\overline{11}$	$\overline{11}$	$\overline{34}$	

Glossary

- $\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3\}$ – set of all integers.
- $\mathbb{Z}^+$ – all positive integers (excludes zero)
- $\mathbb{R}$ – set of all real numbers

Question 1 (12 Marks)

Commence a NEW page.

Marks

- (a) Given that $a^2 + b^2 \geq 2ab$ and $a^2 + b^2 + c^2 \geq ab + bc + ac$, where a, b and $c \in \mathbb{R}^+$, show that:

i. $\sin^2 \alpha + \cos^2 \alpha \geq \sin 2\alpha$. **1**

ii. $\sin^2 \alpha + \cos^2 \alpha + \tan^2 \alpha \geq \sin \alpha - \cos \alpha + \sec \alpha + \frac{1}{2} \sin 2\alpha$. **3**

- (b) i. Show that $\binom{n}{r} < n \binom{n-1}{r-1}$, where $n, r \in \mathbb{Z}^+$ and $1 < r \leq n$. **2**

- ii. Given that $(a+b)^n$ can be written as **2**

$$\binom{n}{0}a^n + \binom{n}{1}a^{n-1}b + \dots + \binom{n}{n}b^n$$

Show that for $a, b > 0$ and $n \in \mathbb{Z}^+$,

$$(a+b)^n - a^n < nb(a+b)^{n-1}$$

- (c) Show that $x > \frac{3 \sin x}{2 + \cos x}$ for $x > 0$. **4**

Question 2 (11 Marks)

Commence a NEW page.

Marks

A particle of mass m kg is dropped from rest in a medium in which the resistance to motion has magnitude $\frac{1}{10}mv^2$ when the velocity of the particle is v ms⁻¹.

After t seconds, the particle has fallen x metres and has velocity v ms⁻¹ and acceleration a ms⁻². Take the acceleration due to gravity as 10 ms⁻².

- (a) Draw a diagram showing the forces acting on the particle. Hence show that **1**

$$a = \frac{100 - v^2}{10}$$

- (b) Show that $t = \frac{1}{2} \log_e \left(\frac{10+v}{10-v} \right)$. **3**

- (c) Find expressions in terms of t for v and x . **4**

- (d) Show that the terminal velocity is 10 ms⁻¹. **1**

- (e) Find the exact time taken and the exact distance fallen by the particle in reaching a speed equal to 80% of its terminal velocity. **2**

Question 3 (11 Marks)

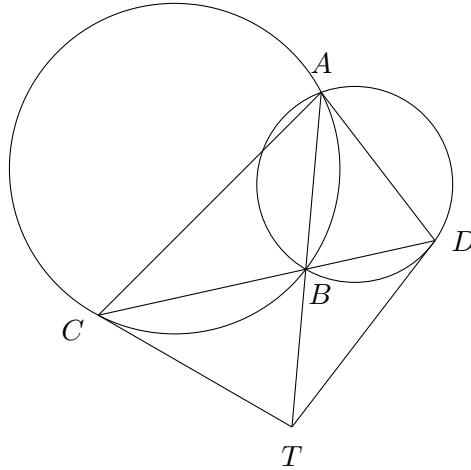
Commence a NEW page.

Marks

- (a) Use mathematical induction to prove for $n \geq 1$, $n \in \mathbb{Z}^+$: **3**

$$1 \times 1! + 2 \times 2! + \cdots + n \times n! = (n+1)! - 1$$

- (b) BAC , BAD are two circles such that tangents at C and D meet at T on AB produced. If CBD is a straight line, prove that



- i. $TC = TD$. **1**
 - ii. $\angle TAC = \angle TAD$. **2**
 - iii. $TCAD$ is a cyclic quadrilateral. **2**
- (c) The number f_n is defined as $f_n = 2^{2^n} + 1$ for $n \in \mathbb{Z}^+$, where 2^{2^n} is 2 raised to the power of 2^n . **3**

Prove, using mathematical induction, that for all $n \in \mathbb{Z}^+$,

$$f_0 f_1 f_2 \cdots f_{n-1} = f_n - 2$$

End of paper.

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1} + C, \quad n \neq -1; \quad x \neq 0 \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x + C, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax} + C, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax + C, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax + C, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax + C, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax + C, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a} + C, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a} + C, \quad a > 0, -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right) + C, \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right) + C$$

NOTE: $\ln x = \log_e x$, $x > 0$

QUESTION 1

(a) (i) let $a = \sin x$ and $b = \cos x$
 then $\sin^2 x + \cos^2 x \geq 2 \sin x \cos x$
 i.e. $\sin^2 x + \cos^2 x \geq \sin 2x$

1 mark
 for correct proof

(ii) let $a = \sin x$, $b = \cos x$, $c = \tan x$
 then $\sin^2 x + \cos^2 x + \tan^2 x \geq \sin x \cos x + \sin x \tan x + \cos x \tan x$

$$\begin{aligned} \text{RHS} &= \frac{1}{2} \sin 2x + \sin x + \frac{\sin^2 x}{\cos x} \\ &= \frac{1}{2} \sin 2x + \sin x + \frac{1 - \cos^2 x}{\cos x} \\ &= \frac{1}{2} \sin 2x + \sin x + \sec x - \cos x \end{aligned}$$

1 mark

~~1 mark~~

1 mark

1 mark

(b) (i)
$$\binom{n}{r} = \frac{n!}{r!(n-r)!} = \frac{n(n-1)!}{r(r-1)!(n-r)!}$$

$$= \frac{1}{r} \times n \times \binom{n-1}{r-1}$$

$$< n \binom{n-1}{r-1} \quad \text{because } r > 1$$

$$\therefore \frac{1}{r} < 1$$

1 mark

1 mark

(ii)
$$\begin{aligned} (a+b)^n - a^n &= \binom{n}{0} a^n + \binom{n}{1} a^{n-1} b + \dots + \binom{n}{n} b^n - a^n \\ &= \binom{n}{1} a^{n-1} b + \binom{n}{2} a^{n-2} b^2 + \dots + \binom{n}{n} b^n \\ &< n \binom{n-1}{0} a^{n-1} b + n \binom{n-1}{1} a^{n-2} b^2 + \dots + n \binom{n-1}{n-1} b^n \\ &\quad \text{(from (i))} \\ &= nb \left[\binom{n-1}{0} a^{n-1} + \binom{n-1}{1} a^{n-2} b + \dots + \binom{n-1}{n-1} b^{n-1} \right] \\ &= nb (a+b)^{n-1} \\ \text{i.e. } (a+b)^n - a^n &< nb (a+b)^{n-1} \end{aligned}$$

1 mark

1 mark

(c) let $f(x) = x - \frac{3 \sin x}{2 + \cos x}$

then $f'(x) = 1 - \frac{(2 + \cos x)3 \cos x - 3 \sin x \cdot -\sin x}{(2 + \cos x)^2}$

$$= 1 - \frac{6 \cos x + 3 \cos^2 x + 3 \sin^2 x}{(2 + \cos x)^2}$$

$$= \frac{4 + 4 \cos x + \cos^2 x - 6 \cos x - 3}{(2 + \cos x)^2}$$

$$= \frac{1 - 2 \cos x + \cos^2 x}{(2 + \cos x)^2}$$

$$= \left(\frac{1 - \cos x}{2 + \cos x} \right)^2$$

$f'(x) \geq 0$ for $x > 0$

$f'(0) = 0$ and $f'(2n\pi) = 0$

$\therefore f(x)$ is non-decreasing for $x > 0$

$f(0) = 0$ and $f(2n\pi) > 0$

$\therefore f(x) > 0$ for $x > 0$

ie $x > \frac{3 \sin x}{2 + \cos x}$ for $x > 0$

1 mark

~~1 mark~~

1 mark

1 mark

1 mark

QUESTION 2

(a)

$$\begin{array}{c} \uparrow \frac{1}{10}mv^2 \\ \downarrow 10m \end{array}$$

$$ma = 10m - \frac{1}{10}mv^2$$

$$a = 10 - \frac{1}{10}v^2 = \frac{100-v^2}{10}$$

1 mark

(b)

$$\frac{dv}{dt} = \frac{100-v^2}{10}$$

$$\frac{dt}{dv} = \frac{10}{100-v^2} = \frac{10}{(10+v)(10-v)}$$

$$= \frac{1}{2} \left(\frac{1}{10+v} + \frac{1}{10-v} \right)$$

using partial fractions

$$\frac{10}{(10+v)(10-v)} = \frac{b}{10+v} + \frac{c}{10-v}$$

$$10 = b(10-v) + c(10+v)$$

$$\text{sub } v=10 \Rightarrow c = \frac{1}{2}$$

$$\text{sub } v=-10 \Rightarrow b = \frac{1}{2}$$

1 mark for reducing to partial fractions

1 mark for integration

1 mark for evaluating constant

1 mark for e^{2t}

1 mark for v

1 mark for $\frac{dx}{dt}$

1 mark for x

1 mark

1 mark for any of these expressions

1 mark for any of these expressions

$$t = \frac{1}{2} (\ln(10+v) - \ln(10-v)) + k$$

$$\text{when } t=0, v=0 \therefore k=0$$

$$\therefore t = \frac{1}{2} \ln \left(\frac{10+v}{10-v} \right)$$

(c)

$$e^{2t} = \frac{10+v}{10-v}$$

$$10e^{2t} - ve^{2t} = 10+v$$

$$v = \frac{10(e^{2t}-1)}{e^{2t}+1}$$

$$\frac{dx}{dt} = \frac{10(e^t - e^{-t})}{e^t + e^{-t}}$$

$$x = 10 \ln(e^t + e^{-t}) + C$$

$$\text{when } t=0, x=0 \therefore C = -10 \ln 2$$

$$\therefore x = 10 \ln \left(\frac{e^t + e^{-t}}{2} \right)$$

(d)

$$\text{as } a \rightarrow 0, \frac{100-v^2}{10} \rightarrow 0$$

$$v^2 = 100$$

$$v = 10 \quad (v > 0)$$

(e)

$$v=8 \Rightarrow t = \frac{1}{2} \ln \left(\frac{18}{2} \right)$$

$$= \frac{1}{2} \ln 9$$

$$= \ln 3$$

$$x = 10 \ln \left(\frac{e^{\ln 3} + e^{-\ln 3}}{2} \right)$$

$$= 10 \ln \left(\frac{3+\frac{1}{3}}{2} \right)$$

$$= 10 \ln \left(\frac{5}{3} \right)$$

QUESTION 3

(a) When $n=1$, LHS = $1 \times 1! = 1$
 RHS = $2! - 1 = 2 - 1 = 1$
 LHS = RHS
 $\therefore$ true for $n=1$

Assume true for $n=k$

i.e. $1 \times 1! + 2 \times 2! + \dots + k \times k! = (k+1)! - 1$

Let $n=k+1$, LHS = $1 \times 1! + 2 \times 2! + \dots + (k \times k! + (k+1) \times (k+1)!)$

$= (k+1)! - 1 + (k+1)(k+1)!$
 from assumption

$= (k+1)! (1 + k+1) - 1$

$= (k+2)(k+1)! - 1$

$= (k+2)! - 1 = \text{RHS}$

i.e. true for $n=k+1$ if true for $n=k$

But proved true for $n=1$

$\therefore$ true for $n=2, 3, 4, \dots$ and all positive integers

(b) (i) $TC^2 = TB \cdot TA$ (Square of tangent equals product of intercepts of secant from the same point)
 $TD^2 = TB \cdot TA$
 $\therefore TC^2 = TD^2$
 $\therefore TC = TD$

(ii) Then $\angle TCD = \angle TDC$ (isos. ΔTCD)
 $\angle TCD = \angle TAC$ (angle between tangent and chord)
 and $\angle TDC = \angle TAD$ (equals angle in alternate segment)
 $\therefore \angle TAC = \angle TAD$

(iii) From (ii) $\angle TCD = \angle TAC = \angle TAD$
 $\therefore TCAD$ is a cyclic quadrilateral
 as the angles in the same segment at C and A subtended from TD are equal

1 mark for proving true for $n=1$

1 mark for correctly substituting from assumption

1 mark for correct proof for $n=k+1$

1 mark - reason not necessary

2 marks for correct proof

-1 mark for ~~no~~ reasons or incorrect reasons

1 mark

1 mark

(c) When $n=1$, $LHS = f_0 = 2^{2^0} + 1 = 2^1 + 1 = 3$
 $RHS = f_1 - 2 = 2^{2^1} + 1 - 2 = 2^2 + 1 - 2 = 3$
 $LHS = RHS$
 $\therefore$ true for $n=1$

Assume true for $n=k$

i.e. $f_0 f_1 \dots f_{k-1} = f_k - 2$

Let $n=k+1$,

$$\begin{aligned} LHS &= f_0 f_1 \dots f_{k-1} f_k \\ &= (f_k - 2) f_k \quad \text{from assumption} \\ &= (2^{2^k} + 1 - 2)(2^{2^k} + 1) \\ &= (2^{2^k} - 1)(2^{2^k} + 1) \\ &= (2^{2^k})^2 - 1 \\ &= 2^{2^k \times 2} - 1 \\ &= 2^{2^{k+1}} - 1 \\ &= 2^{2^{k+1}} + 1 - 2 = f_{k+1} - 2 \\ &= RHS \end{aligned}$$

i.e. true for $n=k+1$ if true for $n=k$

But true for $n=1$, $\therefore$ true for $n=2, 3, 4, \dots$
 and all positive integers

1 mark for
proving true for $n=1$

2 marks for
correct proof

1 mark for
correct substitution
from assumption